

ACCOUNTING

Unit-III: Cost Accounting

Comprehensive Study Notes

UNIT-III: COST ACCOUNTING

1. Introduction to Cost Accounting

1.1 What is Cost Accounting?

Cost Accounting is the process of recording, classifying, analyzing, summarizing, and allocating costs to determine the cost of products or services.

Wheldon: "Cost accounting is the process of accounting for costs from the point at which expenditure is incurred to the point at which it is charged to cost units."

ICMA: "Cost accounting is a branch of accounting that deals with the recording, classification, allocation, and reporting of costs."

1.2 Features of Cost Accounting

Feature	Description
Systematic Recording	Systematic recording of all costs
Cost Analysis	Analysis of cost components
Cost Ascertainment	Determining cost of products/services
Cost Control	Controlling costs to maximize profit
Cost Reduction	Reducing costs without affecting quality
Decision Making	Helps in decision-making
Reporting	Preparing reports for management

1.3 Scope of Cost Accounting

+-----+	+-----+
COST ASCERTAINMENT	COST CONTROL
+-----+	+-----+
+-----+	+-----+
Cost of Product	Budgetary Control
Cost of Service	Standard Costing
Cost of Job	Variance Analysis
Cost of Process	Cost Reduction
+-----+	+-----+

1.4 Advantages of Cost Accounting

Advantage	Description
Cost Control	Helps in controlling costs
Profit Maximization	Maximizes profit by cost reduction
Decision Making	Provides data for decisions
Pricing	Helps in determining selling price
Efficiency	Evaluates efficiency of operations
Budgeting	Facilitates budgeting
Comparison	Compares actual with standard
Loss Minimization	Minimizes losses

1.5 Limitations of Cost Accounting

Limitation	Description
Expensive	Cost accounting is expensive to maintain
Complex	Complex system to understand
Time-Consuming	Requires time and effort
Estimates	Based on estimates and assumptions
No Uniformity	No uniform system followed
Human Error	Subject to human errors

2. Classification of Costs

2.1 What is Cost?

Cost is the monetary value of resources consumed in producing goods or services.

Definition: "Cost is the amount of expenditure incurred on or attributable to a given thing."

+-----+	+-----+
BY NATURE	BY FUNCTION
(Elements of Cost)	
+-----+	+-----+
+-----+	+-----+
Material	Production Cost
Labour	Administration Cost
Expenses	Selling & Distribution
+-----+	+-----+

2.2 Classification by Nature (Elements of Cost)

2.2.1 Material Cost

Type	Description	Example
Direct Material	Materials directly used in production	Raw materials, Components
Indirect Material	Materials not directly used in production	Lubricants, Consumable stores

2.2.2 Labour Cost

Type	Description	Example
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Direct Labour	Labour directly involved in production	Workers, Machine operators
Indirect Labour	Labour not directly involved in production	Supervisor, Cleaner

2.2.3 Expenses

Type	Description	Example
Direct Expenses	Expenses directly related to production	Royalty, Hire charges
Indirect Expenses	Expenses not directly related to production	Rent, Insurance, Depreciation

2.3 Classification by Function

Function	Description
Production/Manufacturing Cost	Cost incurred in production
Administration Cost	Cost of administration
Selling Cost	Cost incurred in selling
Distribution Cost	Cost incurred in distribution

2.4 Classification by Behavior

2.4.1 Fixed Cost

Feature	Description
Definition	Cost that does not change with change in output
Example	Rent, Insurance, Salaries
Behavior	Remains constant
Control	Period cost

2.4.2 Variable Cost

Feature	Description
Definition	Cost that changes with change in output
Example	Raw material, Direct labour
Behavior	Varies proportionately
Control	Product cost

2.4.3 Semi-Variable Cost

Feature	Description
Definition	Cost that has both fixed and variable elements
Example	Telephone charges, Electricity
Behavior	Varies with output but not proportionately

2.5 Classification by Controllability

Type	Description
Controllable Cost	Cost that can be controlled by management
Uncontrollable Cost	Cost that cannot be controlled

3. Cost Sheet

3.1 What is a Cost Sheet?

Cost Sheet is a statement that shows the various components of total cost for a product or service. It provides a detailed breakdown of all costs incurred.

Definition: "A cost sheet is a statement prepared to show the total cost and the cost per unit of a product."

3.2 Components of Cost Sheet

COST SHEET FORMAT
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Particulars	Amount (?)	Per Unit (?)

Direct Material Consumed	xxx	xxx
Direct Labour	xxx	xxx
Direct Expenses	xxx	xxx

PRIME COST	xxx	xxx
Add: Factory Overheads	xxx	xxx

WORKS/FACTORY COST	xxx	xxx
Add: Administrative Overheads	xxx	xxx

COST OF PRODUCTION	xxx	xxx
Add: Selling & Distribution OH	xxx	xxx

TOTAL COST	xxx	xxx
Add: Profit	xxx	xxx

SALES	xxx	xxx

3.3 Solved Example: Cost Sheet

Example

Prepare a cost sheet from the following data:

Particulars	Amount (?)
Opening Stock of Raw Materials	20,000
Purchases of Raw Materials	1,00,000
Closing Stock of Raw Materials	25,000
Direct Wages	40,000
Direct Expenses	10,000
Factory Expenses	15,000
Administrative Expenses	12,000
Selling Expenses	8,000
Number of units produced	1,000

3.4 Practice Questions

Q.No.	Question	Total Cost
Q1	Prepare cost sheet: Material Consumed ₹1,50,000; Wages ₹50,000; Direct Expenses ₹15,000; Factory Expenses ₹20,000; Administrative Expenses ₹18,000; Selling Expenses ₹12,000; Units Produced 2,000	₹2,65,000
Q2	Prepare cost sheet: Opening Stock ₹30,000; Purchases ₹1,20,000; Closing Stock ₹35,000; Wages ₹60,000; Factory Expenses ₹25,000; Admin Expenses ₹15,000; Selling Expenses ₹10,000; Units 1,500	₹2,25,000

4. Marginal Costing

4.1 What is Marginal Costing?

Marginal Costing is a costing technique where only variable costs are considered in product costing. Fixed costs are treated as period costs.

Definition: "Marginal costing is the ascertainment of marginal costs and of the effect on profit of changes in volume or type of output."

4.2 Key Terms in Marginal Costing

Term	Definition	Formula
Contribution	Excess of sales over variable cost	Sales - Variable Cost
Profit Volume Ratio (PV Ratio)	Contribution as percentage of sales	$(\text{Contribution} / \text{Sales}) \times 100$
Break-Even Point (BEP)	Point of no profit no loss	Fixed Cost / Contribution per unit
Margin of Safety	Excess of actual sales over BEP sales	Actual Sales - BEP Sales

Marginal Costing Formulas

Contribution	= Sales - Variable Cost
PV Ratio	= $(\text{Contribution} / \text{Sales}) \times 100$
BEP (units)	= Fixed Cost / Contribution per unit
BEP (?)	= Fixed Cost / PV Ratio
Margin of Safety	= Actual Sales - BEP Sales
Margin of Safety %	= $(\text{Margin of Safety} / \text{Actual Sales}) \times 100$

4.3 Break-Even Point (BEP)

Break-Even Point is the level of sales where total revenue equals total cost (no profit, no loss).

4.4 Solved Examples

Example 1: Calculate Contribution and BEP

Given: Sales ₹5,00,000; Variable Cost ₹3,00,000; Fixed Cost ₹1,00,000

Contribution = Sales - Variable Cost = ₹5,00,000 - ₹3,00,000 = ₹2,00,000
 PV Ratio = (Contribution / Sales) × 100 = (₹2,00,000 / ₹5,00,000) × 100 = 40%
 BEP (₹) = Fixed Cost / PV Ratio = ₹1,00,000 / 0.40 = ₹2,50,000

Example 2: Calculate Margin of Safety

Given: Actual Sales ₹4,00,000; BEP Sales ₹2,00,000

Margin of Safety = Actual Sales - BEP Sales = ₹4,00,000 - ₹2,00,000 = ₹2,00,000
 Margin of Safety % = (Margin of Safety / Actual Sales) × 100 = (₹2,00,000 / ₹4,00,000) × 100 = 50%

Example 3: Find BEP using Contribution per unit

Given: Selling Price per unit ₹50; Variable Cost per unit ₹30; Fixed Cost ₹1,00,000

Contribution per unit = SP - VC = ₹50 - ₹30 = ₹20
 BEP (units) = Fixed Cost / Contribution per unit = ₹1,00,000 / ₹20 = 5,000 units
 BEP (₹) = BEP units × SP = 5,000 × ₹50 = ₹2,50,000

4.5 Advantages of Marginal Costing

Advantage	Description
Simple	Simple to understand and apply
Profit Planning	Useful for profit planning
Decision Making	Helps in short-term decisions
Pricing	Helps in pricing decisions
Cost Control	Useful for cost control
Inventory Valuation	Simple inventory valuation

4.6 Disadvantages of Marginal Costing

Disadvantage	Description
Long-term Pricing	Not useful for long-term pricing
Fixed Cost Ignored	Fixed costs are ignored in product cost
Inventory Valuation	Not acceptable for financial reporting
Complex	May be complex in practice

5. CVP Analysis

5.1 What is CVP Analysis?

Cost Volume Profit (CVP) Analysis examines the relationship between costs, volume, and profit. It helps in decision-making by showing how changes in costs and volume affect profits.

5.2 Key Assumptions of CVP Analysis

Assumption	Description
1	Selling price remains constant
2	Variable cost per unit remains constant
3	Fixed cost remains constant
4	Sales mix remains constant (for multiple products)
5	Production equals sales

5.3 Applications of CVP Analysis

Application	Description
Profit Planning	Planning profit levels
Break-Even Analysis	Determining break-even point
Pricing Decisions	Setting selling price
Make or Buy Decisions	Decide whether to make or buy
Sales Mix	Optimal sales mix decisions

5.4 Solved Examples

Example 1: CVP Analysis - Profit Calculation

Given: Selling Price ₹100; Variable Cost ₹60; Fixed Cost ₹2,00,000; Units Sold 8,000

Contribution per unit = ₹100 - ₹60 = ₹40
 Total Contribution = ₹40 × 8,000 = ₹3,20,000
 Profit = Contribution - Fixed Cost
 = ₹3,20,000 - ₹2,00,000 = ₹1,20,000

Example 2: Desired Profit Calculation

Given: SP ₹50; VC ₹30; FC ₹1,00,000; Desired Profit ₹50,000

Contribution per unit = ₹50 - ₹30 = ₹20
 Required Contribution = FC + Desired Profit = ₹1,00,000 + ₹50,000 = ₹1,50,000
 Required Units = Required Contribution / Contribution per unit
 = ₹1,50,000 / ₹20 = 7,500 units
 Required Sales = 7,500 × ₹50 = ₹3,75,000

5.5 Practice Questions

Q.No. Question

Q1	Calculate Contribution, PV Ratio, and BEP from: Sales ₹6,00,000; Variable Cost ₹3,60,000; Fixed Cost ₹1,20,000
Q2	Calculate BEP in units and value: SP ₹40; VC ₹25; FC ₹90,000
Q3	Calculate Margin of Safety: Actual Sales ₹5,00,000; BEP Sales ₹3,00,000

6. Questions

Short Answer Questions (2-5 Marks)

1. What is cost accounting?
2. What is the nature of cost accounting?
3. What is a cost sheet?
4. What are the elements of cost?
5. What is marginal costing?
6. What is break-even point?
7. What is CVP analysis?
8. What is contribution?
9. What is PV ratio?
10. What is margin of safety?

Long Answer Questions (10 Marks)

1. Explain the nature and scope of cost accounting.
2. Explain the classification of costs.
3. Prepare a cost sheet with examples.
4. Explain marginal costing with its features.
5. Explain CVP analysis and its applications.

Practice Questions

Q.No.	Question
Q1	Prepare a cost sheet with given data
Q2	Calculate BEP and margin of safety
Q3	Calculate contribution, PV ratio, and BEP
Q4	Prepare cost sheet and find per unit cost

Objective Questions (1 Mark)

1. Direct material + Direct Labour =:
(a) Prime Cost (b) Works Cost (c) Total Cost (d) None
2. Contribution =:
(a) Sales - Variable Cost (b) Sales - Fixed Cost (c) Total Cost - Sales (d) None
3. BEP formula is:
(a) $FC/Contribution \text{ per unit}$ (b) $VC/Contribution \text{ per unit}$ (c) $Sales/Contribution$ (d) None
4. Fixed cost + Variable cost =:
(a) Total Cost (b) Prime Cost (c) Works Cost (d) None
5. PV Ratio formula:
(a) $Contribution/Sales \times 100$ (b) $Sales/Contribution \times 100$ (c) $Fixed \text{ Cost}/Sales \times 100$ (d) None

7. Quick Revision

Cost Classification Summary

COST CLASSIFICATION

By Element:	Material + Labour + Expenses
By Nature:	Direct / Indirect
By Function:	Production / Administration / Selling & Distribution
By Behavior:	Fixed / Variable / Semi-Variable
By Time:	Historical / Predetermined
By Control:	Controllable / Uncontrollable

Cost Sheet Components

Prime Cost	= Direct Material + Direct Labour + Direct Expenses
Works Cost	= Prime Cost + Factory Overheads
Cost of Production	= Works Cost + Administrative Overheads
Total Cost	= Cost of Production + Selling & Distribution Overheads

Marginal Costing Formulas

Contribution	= Sales - Variable Cost
PV Ratio	= (Contribution / Sales) x 100
BEP (units)	= Fixed Cost / Contribution per unit
BEP (?)	= Fixed Cost / PV Ratio
Margin of Safety	= Actual Sales - BEP Sales

End of Unit-III Notes